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ALGORITHMIC PRIMITIVES AND COMPOSITIONAL GEOMETRY OF REASONING IN LANGUAGE MODELS

A PREPRINT

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ABSTRACT

How do latent and inference time computations enable large language models (LLMs) to solve multi-step reasoning? We introduce a framework for tracing and steering algorithmic primitives that underlie model reasoning. Our approach links reasoning traces to internal activations and evaluates algorithmic primitives by injecting them into residual streams and measuring their effect on reasoning steps and task performance. We consider four benchmarks: Traveling Salesperson Problem (TSP), 3SAT, AIME, and graph navigation. We operationalize primitives by clustering activations and annotating their matched reasoning traces using an automated LLM pipeline. We then apply function vector methods to derive primitive vectors as reusable compositional building blocks of reasoning. Primitive vectors can be combined through addition, subtraction, and scalar operations, revealing a geometric logic in activation space. Cross-task and cross-model evaluations (Phi-4, Phi-4-Reasoning, Llama-3-8B) show both shared and task-specific primitives. Notably, comparing Phi-4 with its reasoning-finetuned variant highlights compositional generalization after finetuning: Phi-4-Reasoning exhibits more systematic use of verification and path-generation primitives. Injecting the associated primitive vectors in Phi-4 induces behavioral hallmarks associated with Phi-4-Reasoning. Together, these findings demonstrate that reasoning in LLMs may be supported by a compositional geometry of algorithmic primitives, that primitives transfer cross-task and cross-model, and that reasoning finetuning strengthens algorithmic generalization across domains.

1 Introduction

Inference time compute has remarkably improved reasoning in large language models (LLMs), and reasoning-finetuned models like Phi-4-Reasoning substantially outperform their base counterparts on reasoning tasks they were not directly trained on (Abdin et al., 2025). However, the extent to which LLMs, especially reasoning-finetuned models, can learn generalized algorithmic capacities remains poorly understood (Eberle et al., 2025). While recent

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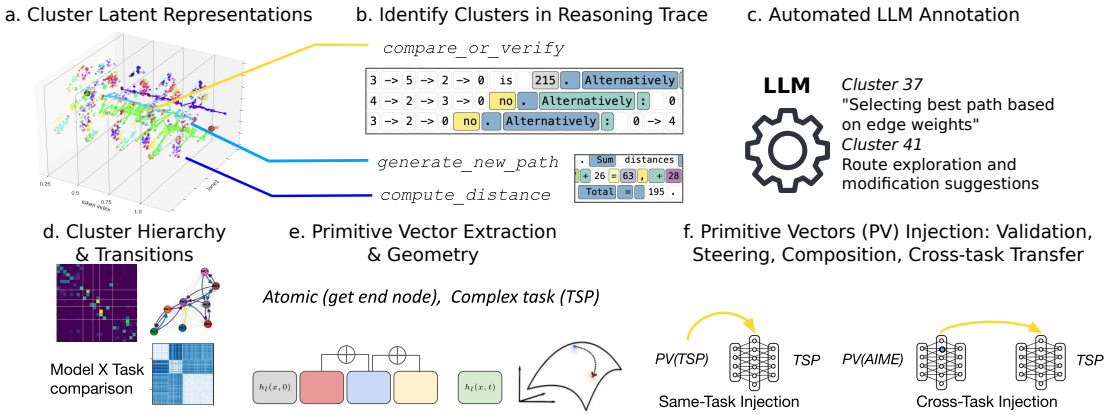

Figure 1: Algorithmic Tracing & Steering: Primitive Extraction & Evaluation. We trace algorithmic primitives by **a** clustering latent representations, **b** identifying corresponding reasoning traces, **c** annotating them using an automated LLM pipeline, **d** meta-clustering primitives, identifying sequential transition trends among clusters, and comparing cluster similarity across models and tasks. Once we identify primitives, **e** we extract associated primitive vectors from top heads, and **f** use causal patching to validate, explore the compositional geometry and cross-task transfer of primitives.

advances in mechanistic interpretability (Todd et al., 2024) point in this direction, it's unclear whether LLMs acquire universal representations of algorithmic primitives (Huh et al., 2024), whether these primitives are geometrically organized in representation space, similar to brains (Fascianelli et al., 2024), and whether LLMs solve reasoning tasks by composition of generalized algorithmic primitives, and through component reuse across tasks and models (Merullo et al., 2024). This presents a unique opportunity to understand the algorithmic basis of LLM reasoning, improvements in compositional generalization after finetuning, and the impact of finetuning on chain-of-thought reasoning (Lobo et al., 2025).

This work aims to understand how fundamental algorithmic primitives are generalized and composed to enable complex reasoning capabilities in LLMs. The work addresses three fundamental questions: (1) What are the basic algorithmic primitives that language models use in specific tasks, and across reasoning domains? (2) Do these primitives compose geometrically in neural activation space? (3) How do reasoning-enhanced models differ from base models in their primitive usage and composition? We introduce a framework for a multi-level algorithmic understanding. We first extract algorithmic primitives by clustering internal representations and leveraging a novel LLM-based automated pipeline for annotating reasoning traces. We then apply function vector methods to extract primitive vectors from the models' internal representations. We induce and steer algorithmic primitives by injecting primitive vectors across the layers. To identify both task-specific and universal algorithmic primitives, we apply the approach across domains: Traveling Salesperson Problem, 3-SAT, AIME, and graph navigation. Finally, to investigate compositional generalization as a result of finetuning, we compare algorithmic performance of base models and their reasoning-finetuned counterparts, like Phi-4 and Phi4-reasoning (Abdin et al., 2025). This multi-level framework allows us to understand the geometry and compositionality of algorithmic primitives in LLM reasoning.

Our contributions include: (1) the systematic identification of cross-domain algorithmic primitives in LLMs, (2) a geometric framework for understanding primitive composition through vector arithmetic, (3) novel methodology linking explicit reasoning behaviors to internal mechanisms, (4) mechanistic evidence for compositional generalization following finetuning LLMs for reasoning, and (5) evaluation of cross-task primitive transfer and generalization.

2 Related Work

Interpretability for Transformers. With increasing scale and complexity of LLMs, the challenge of understanding their predictions has become an important research direction in explainable AI and interpretability research. This has specifically targeted the analysis of internal model structure and feature relationships (Geiger et al., 2022; Eberle et al., 2022; Schnake et al., 2022), as well as representations and manifolds (Kornblith et al., 2019) including concepts (Chormai et al., 2024). Mechanistic interpretability (Sharkey et al., 2025) has focused on the extraction of circuits (Olah et al., 2020; Wang et al., 2023), feature descriptions (Hernandez et al., 2022), and causally effective representations such as function vectors (Todd et al., 2024). Combined analysis of individual neurons (Gurnee & Tegmark, 2024;

Templeton et al., 2024), attention scores (Voita et al., 2019; Clark et al., 2019), and task-specific attention heads (Vig & Belinkov, 2019; McDougall et al., 2024) have further deepened our understanding of internal model processing. In parallel, free-text and chain-of-thought explanations (Turpin et al., 2023; Camburu et al., 2018; Huang et al., 2023; Madsen et al., 2024) consider the model's own natural language explanation of what it is doing and why. Gradient-based feature attributions have further enabled scaling the localization and analysis of relevant features in LLMs (Ali et al., 2022; Jafari et al., 2024).

Function Vectors and Behavioral Interventions. LLMs' contextualized representations such as function vectors (Todd et al., 2024) and in-context task vectors (Hendel et al., 2023; Yang et al., 2025) are shown to trigger execution of associated tasks. A broader application of such steering vectors include persona vectors (Chen et al., 2025) and representations of truthfulness (Marks & Tegmark, 2024; Bürger et al., 2024). Probing has identified associations between internal representations and features of interest (Conneau et al., 2018; Hewitt & Manning, 2019). This has also been extended to the analysis of reasoning strategies, e.g., by identifying probes involved in solving grade-school math problems (Ye et al., 2025). These analyses provide some first methods to understand how complex representational geometries and manifolds drive predictions in modern models. Related work from neuroscience shows that task-related variables are encoded in neural geometries in a format that supports generalization to novel situations (Bernardi et al., 2020). Monkey neuroscience research has linked compositional generalization in neural representational geometry with behavioral performance (Fascianelli et al., 2024). These findings suggest that function vector methods can be used to study how compositional generalization may guide algorithmic steering of LLMs, providing a conceptual starting point for our work.

Algorithmic Evaluation and Steering of Language Models. Multi-step planning and graph navigation are standard benchmarks for structured reasoning (Fatemi et al., 2023), yet most LLMs perform poorly and fail to generalize with growing graph complexity (Momennejad et al., 2023). Although efficient algorithms exist, recent evaluations suggest that LLMs rely on policy-dependent heuristics rather than explicit search strategies (Eberle et al., 2025). Complementary works have aimed to map a high-level causal model to an internal realization by an LLM (Geiger et al., 2024), and prompt models to generate sets of abstract hypotheses about the tasks to improve inductive reasoning performance (Wang et al., 2024). Recent analysis of chain-of-thought outputs has uncovered sentence-level relationships, offering the localization of salient reasoning steps (Bogdan et al., 2025). Existing work either targets high-level behaviors such as politeness, creativity, or honesty using relatively coarse vector representations to steer behavior (Chen et al., 2025), or focuses on very simple input-output mapping (Todd et al., 2024). Our work focuses on algorithmic steering, targeting computational primitives and their compositions, aiming at a deep understanding of the model's internal algorithmic phenotypes and applications in future algorithmic finetuning.

3 Experimental Setup: Models and Tasks

Models. We evaluated decoder-only Transformers on multi-step reasoning tasks. Our analyses focused on Phi-4-Base (Abdin et al., 2024) and the reasoning-specialized variant Phi-4-Reasoning (Abdin et al., 2025), with additional validation from Llama-3-8B (Dubey et al., 2024).

Tasks and Data Collection. Four multi-step reasoning setups were tested. We primarily focused on the Traveling Salesperson Problem (TSP), an NP-hard benchmark. Notably, Phi-4-Reasoning exhibited improved performance on TSP despite not being finetuned for this task (Abdin et al., 2025). We also investigated the 3-SAT and AIME Mathematical Olympiad benchmarks for the same reasons. Finally, we investigated Graph Navigation tasks used in previous LLM reasoning research (Momennejad et al., 2023; Eberle et al., 2025) (see App. B for task prompts). Our setups demand planning, search, and verification. TSP and Graph navigation require graph optimization via planning, path generation, search, comparison, and verification. 3-SAT presents another NP-hard problem that tests flexible algorithmic problem solving. Finally, AIME requires complex mathematical reasoning and multi-step computation. Together, these setups enable us to examine algorithmic primitives that are task-specific and those that generalize across tasks. We collected reasoning traces and associated residual stream vectors across 100+ examples per task, with systematic variation in problem complexity.

4 Methodology

4.1 Definitions and Notation

Definition: Algorithmic Primitive. We define *algorithmic primitive* as a minimal computational operation observed in a reasoning process (Eberle et al., 2025) (e.g. TSP), such as retrieving the nearest neighbor or computing a distance. Primitives can be identified both in explicit reasoning traces (e.g. reasoning steps the model produces in the output) and in internal activations (e.g. clusters of token representations, or attention patterns).